

Summer term 2026

Visual Data Analysis

Assignment Sheet 2

Solution has to be submitted via eCampus by April 24, 2025, 10:00 a.m.

If you have questions concerning the exercises, please use the forum on eCampus.

- Please work on this exercise in **small groups** of 3 students. Submit each solution only once, but clearly indicate who contributed to it by forming a team in eCampus. Remember that all team members have to be able to explain all answers.
- Please submit your answers in PDF format, and your scripts as *.py/*.ipynb files. If you are using [Jupyter notebook](#), please also export your scripts and results as PDF.

Exercise 1 (Communicating with Interactive Articles, 12 Points)

Visualizations are a traditional part of articles in science, education, or journalism. Web-based technologies have made it possible to enhance static text and images with *interactive* visualizations. The paper [Communicating with Interactive Articles](#) gives an overview of such efforts and is, itself, an example of an interactive article.

Note: The goal of this task, and of similar reading tasks on future sheets, is to train your ability to efficiently extract relevant information from articles, and to encourage your own reflection. We select articles that we hope you will find interesting, but we do not expect that you read every word of them, unless you really find them captivating. To obtain the credits, please answer our questions in your own words.

- a) The authors link to many examples of interactive articles. Pick at least one of them, based on your own interest in either the topic or the used visualizations / interactions, and explain your choice in one sentence. Describe at least one interaction that is offered by the article of your choice. What did you perceive as its main benefit? Do you see an opportunity for further improving the visualization or interaction? (5P)
- b) The authors repeatedly highlight findings from psychological and educational research. State two such findings, and explain why they are relevant to interactive articles. (4P)
- c) Even though The New York Times pioneered the integration of interactive visualizations into journalism, it has now become more cautious about the use of interaction. What are reasons for this? Do you think those reasons carry over to other domains, such as science and education? (3P)

Exercise 2 (Plotting Categorical Data, 7 Points)

Please use pandas and seaborn to inspect the numerical attributes of a chronic kidney disease dataset, which is available as an Excel file on eCampus.

- a) Read the dataset using pandas' `read_excel` function. Use `pandas.melt` to transform the data from “wide” to “long” format, using `class` (indicating `ckd` for chronic kidney disease or `notckd` for its absence) as the identifier variable. (2P)
- b) For each numerical attribute, such as age or blood pressure, create two boxplots side-by-side. One should show the attribute's distribution among patients suffering from chronic kidney disease, the other one from patients who do not suffer from the disease. *Hint:* Due to the different numerical ranges, you will have to disable sharing of y axes between plots of different attributes. (4P)
- c) Based on viewing the plots, name an attribute that appears to be highly indicative of chronic kidney disease, and one that seems to be mostly unrelated to it. (1P)

Hint: You can find the seaborn documentation at <https://seaborn.pydata.org/tutorial.html>.

Exercise 3 (Multidimensional Data Filtering and Visualization, 26 Points)

In practice, data filtering and visualization often go hand in hand. In this task, you will continue to use pandas and seaborn for these tasks. We will explore how differences in (sensory) wine quality relate to physicochemical quantities, such as amount of residual sugar or acidity. You can find a more detailed description of this dataset at the [UCI Machine Learning Repository](#)

Hint: Most tasks require writing very little, and not very complex code. The main task will be to identify and make proper use of the required functionality in pandas, seaborn, and scikit learn.

- a) Read the data given in `winequality-red.csv`, available from eCampus, and print the first few rows. (1P)
- b) A numerical rating of sensory wine quality is given in the column “quality”. Display the distribution of these scores with a histogram. What is the range of this score in the data? (1P)
- c) Derive a coarser classification of quality into “low”, “medium”, and “high”, by grouping together the two lowest, the intermediate, and the two highest quality scores that occur in the dataset, respectively. Replace the original “quality” column with a new column “quality bin” that contains these labels. (3P)
- d) We would like to investigate differences between high and low quality wines. Therefore, create a filtered data frame in which the medium-quality wines are omitted. (1P)
- e) Visualize all numerical attributes in a scatterplot matrix. Color the two quality levels differently. (1P) *Hint:* Using seaborn, you can create this plot with a single (and simple) line of code.
- f) Based on the visualization, name five attributes that appear to best distinguish between high and low quality. (1P)
- g) Now, use an automated feature selection technique to identify five attributes that distinguish between high and low quality. More specifically, please use the F score from a one-way analysis of variance (ANOVA) to rank the attributes, as implemented in scikit learn's `f_classif`. What are the five best attributes according to this measure? Are they the same as those you identified visually? Create a filtered data frame that only contains the top five attributes, plus the “quality bin”. (4P)
- h) Create a matrix similar to the one in Fig. 1: It should compare the two quality bins with respect to the five top-ranking attributes, using density estimates (on the diagonal), scatterplots (in the upper triangular part), and plots of pairwise linear regression models (in the lower triangular part). *Hint:* It's convenient to use seaborn's `PairGrid` class for this. (4P)
- i) Based on the visualization, which attributes appear to be strongly correlated regardless of quality? For which attributes does the amount of correlation appear to depend on the quality? Does any of the attributes appear to have a multimodal distribution? Point out one or multiple data points that appear to be outliers. (4P)

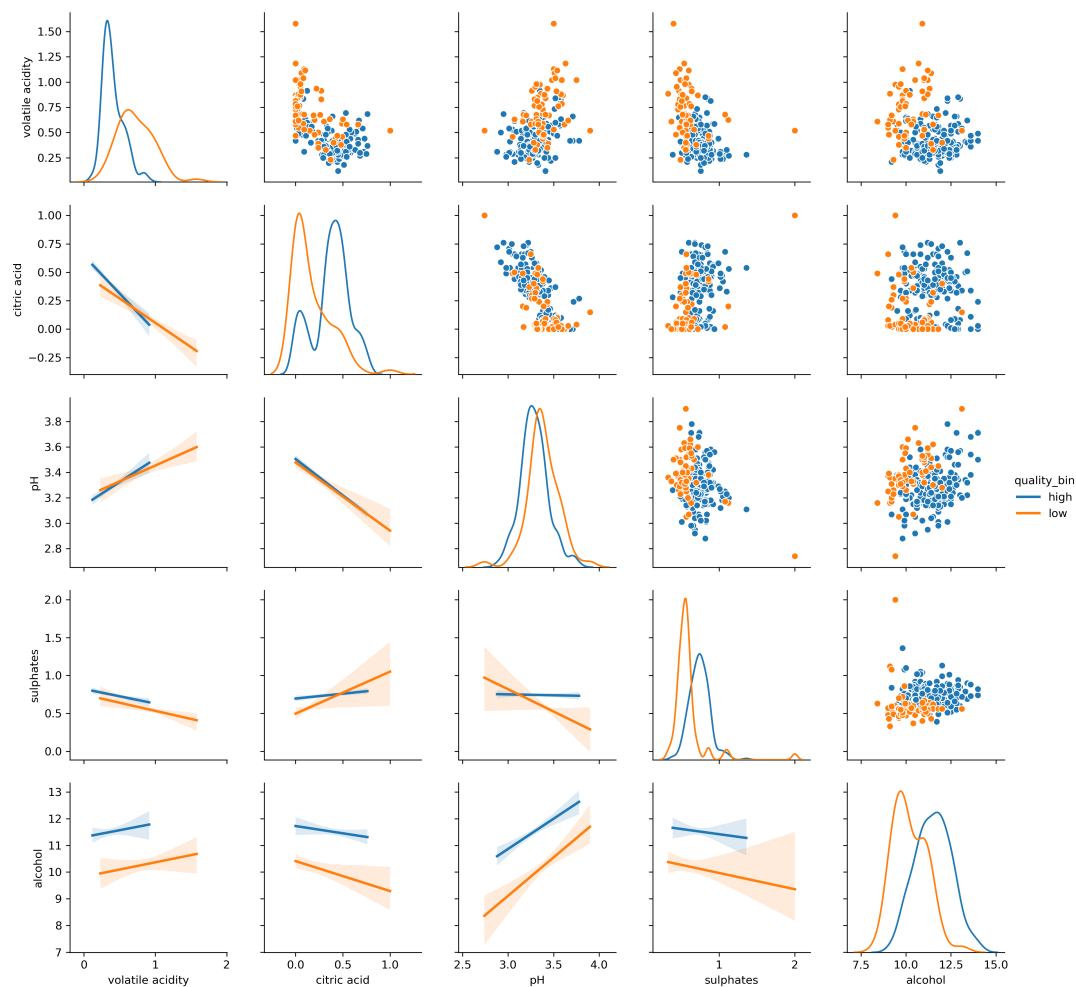


Figure 1: The visualization which you should obtain in step h) of this task.

- j) Compute the distance consistency of all scatter plots. Which pair of variables leads to the highest distance consistency? (6P) *Hint:* To our knowledge, distance consistency is not implemented in any widely used Python package. For this subtask, we expect that you will have to write some code yourself.

Exercise 4 (Kernel Density Estimation, 5 Points)

- The file `values.txt` that is available on eCampus contains 40 scalar values. Use the seaborn library with default parameters to visualize the distribution of those values using Kernel Density Estimation. Based on the visualization, at which location(s) do you think the underlying distribution has a mode (peak)? (2P)
- Produce alternative KDE plots by adjusting the bandwidth to higher and lower values. Briefly describe in your own words how this changes the shape of the estimated distribution. Visualize the same data in a different way to help you decide which setting most faithfully reflects the distribution which generated the data. In particular, at which location(s) would you assume it has modes? (3P)

Good Luck!